Data Profiling Observations & Inconsistencies

During the profiling of the *311 Call Center Service Requests Kansas City (2007–March 2021)* dataset using Alteryx, several issues and observations were noted. These are important to highlight before moving forward with analysis and staging to SQL.

1. Missing Values

- **Address fields**: Several records have missing or null values for Address, Zip Code, or Latitude/Longitude. This impacts geographical visualizations and spatial analysis.
- **Closure/Resolution fields**: Columns like Closed_Date or Resolution_Action contain nulls, especially for cases that are still open or in progress.
- **Contact information fields**: Some optional contact details (e.g., phone/email) are sparsely populated.

2. Duplicates

- Multiple requests appear duplicated with the same Case ID, Service Request Type, and Created Date.
- Duplicates may arise from system re-entry of cases or errors in logging. Deduplication may be necessary for cleaner analysis.

3. Date & Time Format Issues

- Date columns (e.g., Created_Date, Closed_Date) show inconsistencies in format: some are stored as full datetime stamps (with hh:mm:ss), while others only store the date.
- A few records have Closed_Date earlier than Created_Date, indicating data entry or system errors.
- Some records in 2007–2009 contain default values (e.g., 01-01-1900 or NULL), which are placeholders rather than valid entries.

4. Data Type / Conversion Issues

- Numeric fields like Zip Code are treated as numeric in some cases and as text in others, leading to misinterpretation (e.g., loss of leading zeros).
- Latitude and Longitude are sometimes interpreted as averages in Power BI/Tableau due to incorrect data category assignment.
- Response time calculation required conversion between datetime columns, and in some cases missing or invalid dates resulted in negative or null response times.

5. Inconsistent / Dirty Values

- **Department and Work Group names**: Minor variations (e.g., "Public Works" vs "Public_Work") lead to separate categories for the same department.
- **Status field**: Contains multiple variations of the same status ("Closed", "closed", "Closed - Completed"). Requires normalization.
- **Category/Service Type**: Some records have misspellings or truncated values.

6. Outliers

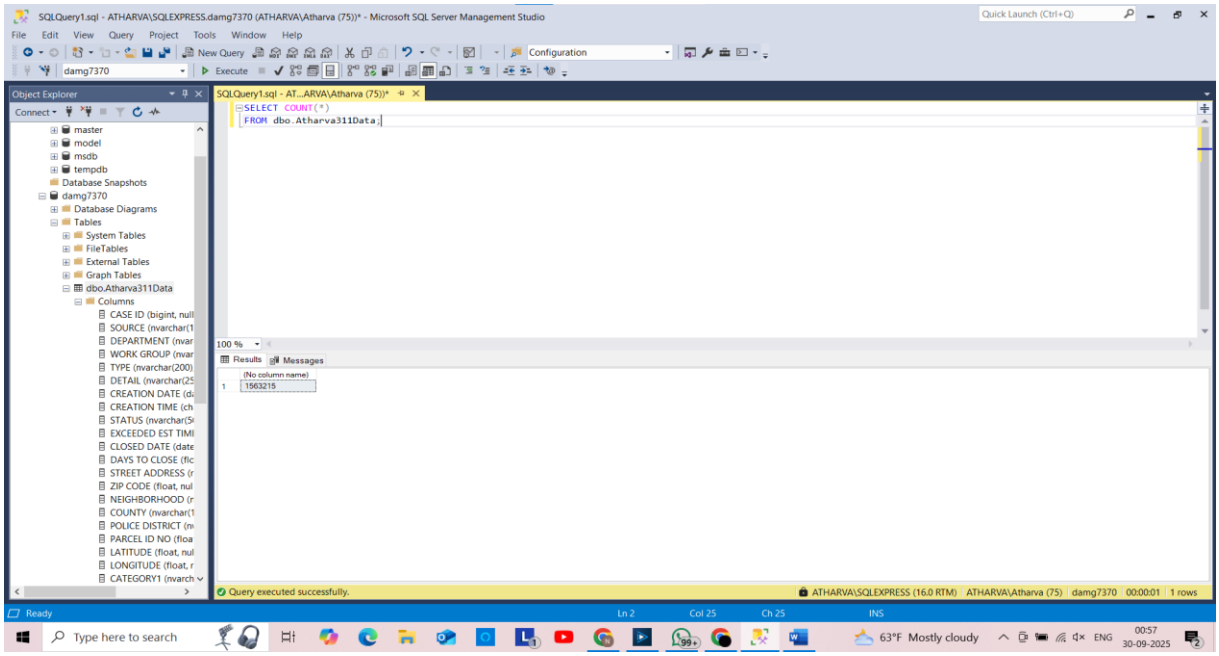
- Response times: A few records show extremely high closure times (e.g., >1000 days), which skew averages and distributions.
- Geographic coordinates: Certain Latitude/Longitude values fall outside Kansas City boundaries, suggesting entry errors.

7. Special Characters & Encoding Issues

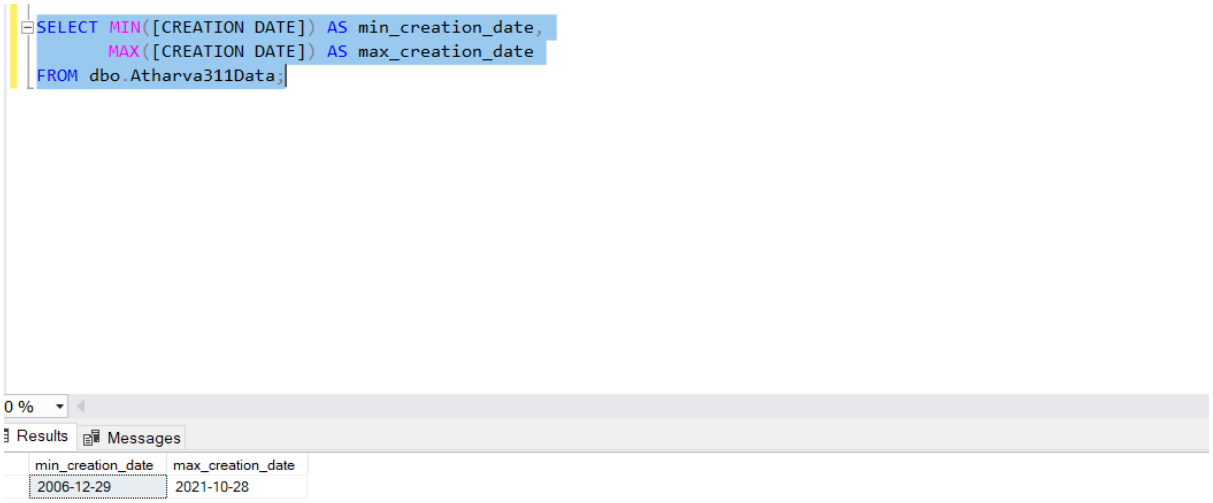
- A few Address and Request Description fields contain special characters (e.g., #, ?, *) that may affect joins or filtering in SQL.
- Non-printable whitespace characters were detected in some text fields, requiring trimming during cleaning.

8. General Observations

- Dataset is very large (2007–2021) and unevenly distributed: request volumes are much higher after 2015 compared to earlier years.
- Seasonal patterns are visible, with spikes in certain request types (e.g., snow/ice removal during winter months).
- A significant number of records are still “Open” or “In Progress,” which may bias trend analysis depending on cut-off date.



Data populated Successfully into MySQL Server.



Quick Check before running Queries.

Q1

A)

```sql
SELECT
    YEAR([CREATION DATE]) AS [Year],
    COUNT(*)              AS Requests
FROM dbo.Atharva311Data
WHERE [CREATION DATE] >= '2018-01-01'
  AND [CREATION DATE] <  '2022-01-01'
GROUP BY YEAR([CREATION DATE])
ORDER BY [Year];
```

Results | Messages

Year	Requests
2018	124280
2019	166021
2020	125906
2021	19683

Q1

B)

```sql
SELECT
    CONVERT(date, DATEFROMPARTS(YEAR([CREATION DATE]),
                                MONTH([CREATION DATE]), 1)) AS [Month],
    COUNT(*) AS Requests
FROM dbo.Atharva311Data
WHERE [CREATION DATE] >= '2018-01-01'
  AND [CREATION DATE] <  '2022-01-01'
GROUP BY CONVERT(date, DATEFROMPARTS(YEAR([CREATION DATE]),
                                     MONTH([CREATION DATE]), 1))
ORDER BY [Month];
```

100 %

Results | Messages

	Month	Requests
1	2018-01-01	10870
2	2018-02-01	7728
3	2018-03-01	9625
4	2018-04-01	9657
5	2018-05-01	12295
6	2018-06-01	11644
7	2018-07-01	12329
8	2018-08-01	10960
9	2018-09-01	9699
10	2018-10-01	10095
11	2018-11-01	11419
12	2018-12-01	7959
13	2019-01-01	20641
14	2019-02-01	15596
15	2019-03-01	15166
16	2019-04-01	16108
17	2019-05-01	16919
18	2019-06-01	15088
19	2019-07-01	14194
20	2019-08-01	13207
21	2019-09-01	10784
22	2019-10-01	10231
23	2019-11-01	7965
24	2019-12-01	10122
25	2020-01-01	12357
26	2020-02-01	11168
27	2020-03-01	10835
28	2020-04-01	9390
29	2020-05-01	11462
30	2020-06-01	12773
31	2020-07-01	12809
32	2020-08-01	10772
33	2020-09-01	10146
34	2020-10-01	8585
35	2020-11-01	7578
36	2020-12-01	8031
37	2021-01-01	7821
38	2021-02-01	8878
39	2021-03-01	2206
40	2021-04-01	649
41	2021-05-01	34
42	2021-06-01	23
43	2021-07-01	14
44	2021-08-01	21
45	2021-09-01	26
46	2021-10-01	11

Q2)

```sql
SELECT [SOURCE], COUNT(*) AS Requests
FROM dbo.Atharva311Data
GROUP BY [SOURCE]
ORDER BY Requests DESC;

-- Monthly by source (for a stacked area/line)
SELECT
    CONVERT(date, DATEFROMPARTS(YEAR([CREATION DATE]),
                                MONTH([CREATION DATE]), 1)) AS [Month],
    [SOURCE],
    COUNT(*) AS Requests
FROM dbo.Atharva311Data
GROUP BY CONVERT(date, DATEFROMPARTS(YEAR([CREATION DATE]),
                                     MONTH([CREATION DATE]), 1)), [SOURCE]
ORDER BY [Month], [SOURCE];
```

100 %

Results | Messages

	SOURCE	Requests
1	PHONE	1204236
2	WEB	211721
3	EMAIL	80585
4	SYS	19226
5	INSPE	14690
6	BOT	13396
7	TWIR	8311
8	VOICE	6021

	Month	SOURCE	Requests
1	2006-12-01	PHONE	1
2	2007-01-01	EIP	50
3	2007-01-01	EMAIL	42
4	2007-01-01	INSPE	1
5	2007-01-01	PHONE	4367
6	2007-01-01	VOICE	8
7	2007-01-01	WEB	73
8	2007-02-01	CTI	1
9	2007-02-01	EIP	7
10	2007-02-01	EMAIL	105

Query executed successfully

Q3)

SQLQuery1.sql - AT...ARVA\Atharva (75))*

```sql
SELECT [DEPARTMENT], COUNT(*) AS Requests
FROM dbo.Atharva311Data
GROUP BY [DEPARTMENT]
ORDER BY Requests DESC;
```

100 %

Results | Messages

	DEPARTMENT	Requests
1	NHS	783094
2	Public Works	353787
3	Water Services	216852
4	Parks and Rec	87954
5	Health	39543
6	KCPD	36369
7	City Managers Office	13098
8	City Planning and Development	12575
9	Northland	8591
10	NCS	6391
11	Finance	1616
12	Parks & Rec	861
13	Fire	612
14	General Service	518
15	Municipal Court	379
16	Housing Community Dev	342
17	South	309
18	Aviation	151
19	Convention and Entertainmen...	59
20	Mayors Office	37
21	Northeast	35
22	Information Technology	24
23	Parks & Recreation	11
24	City Council	3
25	City Clerks Office	2
26	Human Resources	1
27	IT	1

Q4)

```sql
SELECT TOP (10)
    [CASE ID],
    [CATEGORY1],
    [TYPE],
    [CREATION DATE],
    [CLOSED DATE],
    TRY_CONVERT(float, [DAYS TO CLOSE]) AS DaysToClose
FROM dbo.Atharva311Data
WHERE TRY_CONVERT(float, [DAYS TO CLOSE]) IS NOT NULL
ORDER BY TRY_CONVERT(float, [DAYS TO CLOSE]) ASC, [CREATION DATE];
```

100 %

Results | Messages

	CASE ID	CATEGORY1	TYPE	CREATION DATE	CLOSED DATE	DaysToClose
1	2015019840	Snow & Ice	Street	2015-03-02	2015-02-09	-21
2	2015020093	Snow & Ice	Administration	2015-03-02	2015-02-09	-21
3	2015019914	Snow & Ice	Street	2015-03-02	2015-02-09	-21
4	2015019860	Snow & Ice	Street	2015-03-02	2015-02-09	-21
5	2015019919	Snow & Ice	Street	2015-03-02	2015-02-09	-21
6	2015019885	Snow & Ice	Street	2015-03-02	2015-02-09	-21
7	2015019851	Snow & Ice	Street	2015-03-02	2015-02-09	-21
8	2015019845	Snow & Ice	Street	2015-03-02	2015-02-09	-21
9	2015019877	Snow & Ice	Street	2015-03-02	2015-02-09	-21
10	2015019866	Snow & Ice	Street	2015-03-02	2015-02-09	-21

Q5)

By zip code

```sql
SELECT TOP (10)
    CAST(CAST([ZIP CODE] AS int) AS varchar(10)) AS ZipCode,
    COUNT(*) AS Requests
FROM dbo.Atharva311Data
WHERE [ZIP CODE] IS NOT NULL
GROUP BY CAST(CAST([ZIP CODE] AS int) AS varchar(10))
ORDER BY Requests DESC;
```

100 %

Results | Messages

	ZipCode	Requests
1	64130	133154
2	64127	90867
3	64114	83166
4	64134	77097
5	64131	76437
6	64132	76361
7	64128	73073
8	64110	64771
9	64119	57224
10	64111	50195

Q5 by street address

```sql
SELECT TOP (10)
    [STREET ADDRESS],
    COUNT(*) AS Requests
FROM dbo.Atharva311Data
WHERE [STREET ADDRESS] IS NOT NULL AND LTRIM(RTRIM([STREET ADDRESS])) <> ''
GROUP BY [STREET ADDRESS]
ORDER BY Requests DESC;
```

100 %

Results | Messages

	STREET ADDRESS	Requests
1	414 E 12TH ST	5773
2	4800 E 63rd St	2406
3	415 E 12th st	1816
4	400 W 31st St	509
5	5100 Wornall Rd	381
6	4900 Raytown Rd	291
7	2400 Troost Ave	290
8	2700 Main St	278
9	7750 E Front St	269
10	8801 E 47th St	250

Q5) by lat/log

```sql
SELECT TOP (10)
    ROUND(TRY_CONVERT(float, [LATITUDE]), 3)  AS Lat_3dp,
    ROUND(TRY_CONVERT(float, [LONGITUDE]), 3) AS Lon_3dp,
    COUNT(*) AS Requests
FROM dbo.Atharva311Data
WHERE [LATITUDE]  IS NOT NULL
  AND [LONGITUDE] IS NOT NULL
GROUP BY ROUND(TRY_CONVERT(float, [LATITUDE]), 3),
         ROUND(TRY_CONVERT(float, [LONGITUDE]), 3)
ORDER BY Requests DESC;
```

100 %

Results | Messages

	Lat_3dp	Lon_3dp	Requests
1	35.798	-103.891	40610
2	39.1	-94.578	4940
3	39.014	-94.53	2468
4	39.099	-94.578	1265
5	39.286	-94.473	520
6	39.048	-94.572	472
7	39.075	-94.592	468
8	39.308	-94.541	449
9	39.1	-94.577	414
10	39.114	-94.528	402

Q6)

SQLQuery1.sql - AT...ARVA\Atharva (75))*

```sql
SELECT
    [DEPARTMENT],
    [WORK GROUP],
    COUNT(*) AS Requests
FROM dbo.Atharva311Data
GROUP BY [DEPARTMENT], [WORK GROUP]
ORDER BY [DEPARTMENT], Requests DESC;
```

100 %

Results | Messages

	DEPARTMENT	WORK GROUP	Requests
1	Aviation	Aviation--	151
2	City Clerks Office	City Clerks Office--	2
3	City Council	City Council--	3
4	City Managers Office	City Managers Office-311 Call Center-Support	8106
5	City Managers Office	City Managers Office-KCPD-	2131
6	City Managers Office	City Managers Office-311-Field Staff	1235
7	City Managers Office	City Managers Office-CIMO-	579
8	City Managers Office	City Managers Office-311 Call Center-Administratio	397
9	City Managers Office	City Managers Office-Capital Projects-	187
10	City Managers Office	City Managers Office-311 Call Center-	169
11	City Managers Office	City Managers Office-Administration-	132
12	City Managers Office	City Managers Office-Emergency Management-	103
13	City Managers Office	City Managers Office-Compass KC-	59
14	City Planning and Development	City Planning and Development-Permit Compliance-	8170
15	City Planning and Development	City Planning and Development-Dev Comp-Zoning	3507
16	City Planning and Development	City Planning and Development-Adminstration-	377
17	City Planning and Development	City Planning and Development-Land Develop-Inspect	319
18	City Planning and Development	City Planning and Development-Elevators-	185
19	City Planning and Development	City Planning and Development-Administration	17
20	Convention and Entertainmen...	Convention and Entertainment Center--	59
21	Finance	Finance-PIAC-	1611
22	Finance	Finance--	5
23	Fire	Fire--	612
24	General Service	General Service-Facility Management-	452
25	General Service	General Service-City Property-	66
26	Health	Health-Rat-	16760
27	Health	Health-Food Protection-	6616
28	Health	Health-Noise Control-	4101

Query executed successfully. ATHARVA\SQLEXPRESS (16.0 RTM) ATHARVA\Atharva (75) dam

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Q7)

```sql
WITH D AS (
    SELECT
        [DEPARTMENT],
        TRY_CONVERT(float, [DAYS TO CLOSE]) AS DaysToClose
    FROM dbo.Atharva311Data
    WHERE TRY_CONVERT(float, [DAYS TO CLOSE]) IS NOT NULL
)
SELECT DISTINCT
    [DEPARTMENT],
    COUNT(*)                                OVER (PARTITION BY [DEPARTMENT])        AS N,
    AVG(DaysToClose)                        OVER (PARTITION BY [DEPARTMENT])        AS AvgDays,
    MIN(DaysToClose)                        OVER (PARTITION BY [DEPARTMENT])        AS MinDays,
    MAX(DaysToClose)                        OVER (PARTITION BY [DEPARTMENT])        AS MaxDays,
    PERCENTILE_CONT(0.50) WITHIN GROUP (ORDER BY DaysToClose)
        OVER (PARTITION BY [DEPARTMENT])                                            AS P50,
    PERCENTILE_CONT(0.90) WITHIN GROUP (ORDER BY DaysToClose)
        OVER (PARTITION BY [DEPARTMENT])                                            AS P90
FROM D
ORDER BY [DEPARTMENT];
```

100 %

Results | Messages

	DEPARTMENT	N	AvgDays	MinDays	MaxDays	P50	P90
1	Aviation	150	13.0933333333333	0	155	4	35.3
2	City Clerks Office	2	2	2	2	2	2
3	City Council	3	77.6666666666667	69	83	81	82.6
4	City Managers Office	12816	22.1688514357054	-1	2996	5	48
5	City Planning and Development	12023	105.12525991849	0	3595	28	302
6	Convention and Entertainment Center	59	16.4576271186441	0	432	4	19.6
7	Finance	1615	10.4402476780186	0	429	1	25
8	Fire	610	7.63934426229508	0	129	4	16
9	General Service	513	57.5438596491228	0	621	8	182.6
10	Health	39356	11.1741284683403	0	3170	7	24
11	Housing Community Dev	342	1.73391812865497	0	137	0	4
12	Human Resources	1	10	10	10	10	10
13	Information Technology	23	26.5217391304348	0	258	1	102.6
14	IT	1	47	47	47	47	47
15	KCPD	36345	3.62778924198652	0	412	3	6

16	Mayors Office	37	129.486486486486	1	301	144	212.6
17	Municipal Court	379	4.22955145118734	0	99	1	13
18	NCS	6383	767.128779570735	0	1839	745	1804
19	NHS	762...	77.7277530706217	0	4525	5	217
20	Northeast	33	45.7878787878788	1	296	23	108.6
21	Northland	8445	27.5583185316755	0	1182	21	51
22	Parks & Rec	861	43.9059233449477	0	912	13	96
23	Parks & Recreation	11	45.5454545454545	4	117	45	112
24	Parks and Rec	87800	19.9493849658314	0	744	10	48
25	Public Works	352...	10.9746796706317	-21	1319	3	29
26	South	304	44.5986842105263	0	386	28...	111.1
27	Water Services	212...	32.4879225011505	0	1476	5	88

Q8)

```sql
WITH Base AS (
  SELECT
      YEAR([CREATION DATE]) AS [Year],
      [STATUS]              AS Status,
      COUNT(*)              AS Requests
  FROM dbo.Atharva311Data
  WHERE [CREATION DATE] >= '2018-01-01'
    AND [CREATION DATE] <  '2022-01-01'
  GROUP BY YEAR([CREATION DATE]), [STATUS]
),
Totals AS (
  SELECT [Year], SUM(Requests) AS YearTotal
  FROM Base
  GROUP BY [Year]
)
SELECT b.[Year], b.Status, b.Requests,
       CAST(100.0 * b.Requests / t.YearTotal AS decimal(5,2)) AS PctOfYear
FROM Base b
JOIN Totals t ON t.[Year] = b.[Year]
ORDER BY b.[Year], b.Status;
```

100 %

Results | Messages

	Year	Status	Requests	PctOfYear
1	2018	CANC	637	0.51
2	2018	DUP	650	0.52
3	2018	FAIL	1	0.00
4	2018	OPEN	1218	0.98
5	2018	RESOL	121774	97.98
6	2019	ASSIG	1	0.00
7	2019	CANC	432	0.26
8	2019	DUP	171	0.10
9	2019	FAIL	1	0.00
10	2019	OPEN	2229	1.34
11	2019	RESOL	163187	98.29
12	2020	ASSIG	37	0.03

13	2020	CANC	291	0.23
14	2020	DUP	278	0.22
15	2020	OPEN	5820	4.62
16	2020	RESOL	119480	94.90
17	2021	ASSIG	7	0.04
18	2021	CANC	73	0.37
19	2021	DUP	1	0.01
20	2021	OPEN	1343	6.82
21	2021	RESOL	18259	92.77

Q9)

```sql
SELECT TOP (10)
    [CATEGORY1],
    COUNT(*)                                        AS Requests,
    AVG(TRY_CONVERT(float, [DAYS TO CLOSE]))        AS AvgDaysToClose
FROM dbo.Atharva311Data
WHERE [CATEGORY1] IS NOT NULL
  AND TRY_CONVERT(float, [DAYS TO CLOSE]) IS NOT NULL
GROUP BY [CATEGORY1]
ORDER BY AvgDaysToClose DESC;
```

100 %

Results | Messages

	CATEGORY1	Requests	AvgDaysToClose
1	Data Not Available	749	1189.01602136182
2	Weeds	12398	420.305371834167
3	Property & Nuisance Violations	20447	187.382354379616
4	Property Violations	133543	180.22246766959
5	Water Main Break	99	172.50505050505
6	Property	69038	171.236130826501
7	Mowing	33733	162.946076542258
8	Information Request	4	103.5
9	Water Services	5942	99.108044429485
10	Nuisance Violations	35697	93.2275541362019

Q10)

```sql
SELECT
    [DEPARTMENT],
    COUNT(*)                                    AS Requests,
    AVG(TRY_CONVERT(float, [DAYS TO CLOSE]))    AS AvgDaysToClose
FROM dbo.Atharva311Data
WHERE TRY_CONVERT(float, [DAYS TO CLOSE]) IS NOT NULL
GROUP BY [DEPARTMENT]
ORDER BY Requests DESC;
```

100 %

Results | Messages

	DEPARTMENT	Requests	AvgDaysToClose
1	NHS	762712	77.7277530706217
2	Public Works	352918	10.9746796706317
3	Water Services	212958	32.4879225011505
4	Parks and Rec	87800	19.9493849658314
5	Health	39356	11.1741284683403
6	KCPD	36345	3.62778924198652
7	City Managers Office	12816	22.1688514357054
8	City Planning and ...	12023	105.12525991849
9	Northland	8445	27.5583185316755
10	NCS	6383	767.128779570735
11	Finance	1615	10.4402476780186
12	Parks & Rec	861	43.9059233449477
13	Fire	610	7.63934426229508
14	General Service	513	57.5438596491228
15	Municipal Court	379	4.22955145118734
16	Housing Communi...	342	1.73391812865497
17	South	304	44.5986842105263
18	Aviation	150	13.0933333333333
19	Convention and En...	59	16.4576271186441
20	Mayors Office	37	129.486486486486
21	Northeast	33	45.7878787878788
22	Information Techn...	23	26.5217391304348
23	Parks & Recreation	11	45.5454545454545
24	City Council	3	77.6666666666667
25	City Clerks Office	2	2
26	Human Resources	1	10